# CPR iMCQ 4 F25

BCHM/HCB

# CPR IMCQ 4 Fall 2025

Biochemistry 6 Questions

1. A 42-year-old woman comes to the physician because of a 2-day history of pain, swelling, and redness of her right leg. She recently returned from a 14-hour international flight. She has no significant medical history and is not under any medications. Her mother had a history of “blood clots” in her legs during pregnancy. Physical examination shows the right calf is swollen, erythematous, and tender to palpation. Doppler ultrasonography of the right lower extremity shows evidence of a deep vein thrombosis (DVT). Targeted genetic analysis shows a pathogenic loss-of-function mutation in the gene encoding Protein C. Inactivation of which of the following proteins is most likely defective in this patient?

A. Factor III (tissue factor)
B. Factor Va
C. Factor VIIa
D. Factor XIIa
E. Plasmin
F. Factor IIa
G. Antithrombin-III

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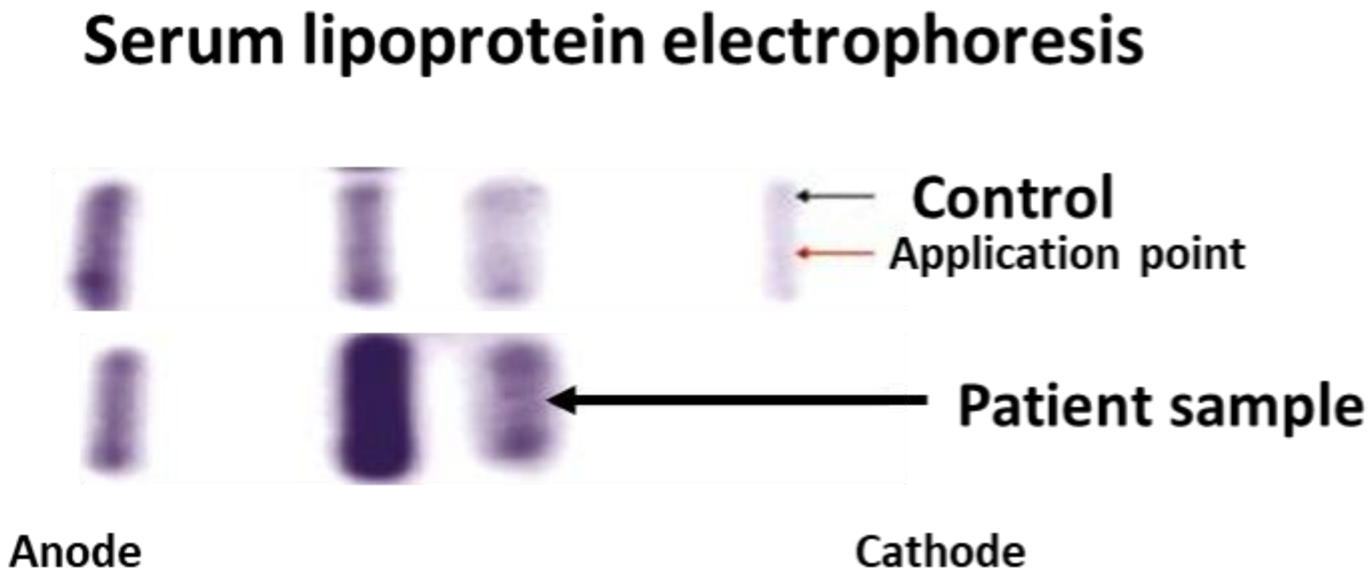
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0258 Outline the anticoagulant and fibrinolytic roles of PGI2, antithrombin III, proteins C and S, and plasmin.

2. A 38-year-old woman comes to the physician for a follow-up examination. She has a 5-year history of type 2 diabetes mellitus and takes oral hypoglycemic agents. She is 167.6 cm (5 ft 6 in) tall, weighs 95.5 kg (210 lb); BMI is 34 kg/$m^2$. Laboratory studies show HbA1c level is 9.7% (normal:<5.7%). Serum lipid studies show a lipemic serum, total cholesterol concentration of 180 mg/dL (normal: <200 mg/dL), and triglyceride concentration of 450 mg/dL (normal: <150 mg/dL). Serum lipoprotein electrophoresis of the patient and control are shown in the figure. Which of the following best explains the lipid abnormality in this patient?

A. Hyperlipidemia type I
B. Hyperlipidemia type IIa
C. Hyperlipidemia type IIb
D. Hyperlipidemia type III
E. Hyperlipidemia type IV
F. Hyperlipidemia type V



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D. Hyperlipidemia type III
✓ E. Hyperlipidemia type IV
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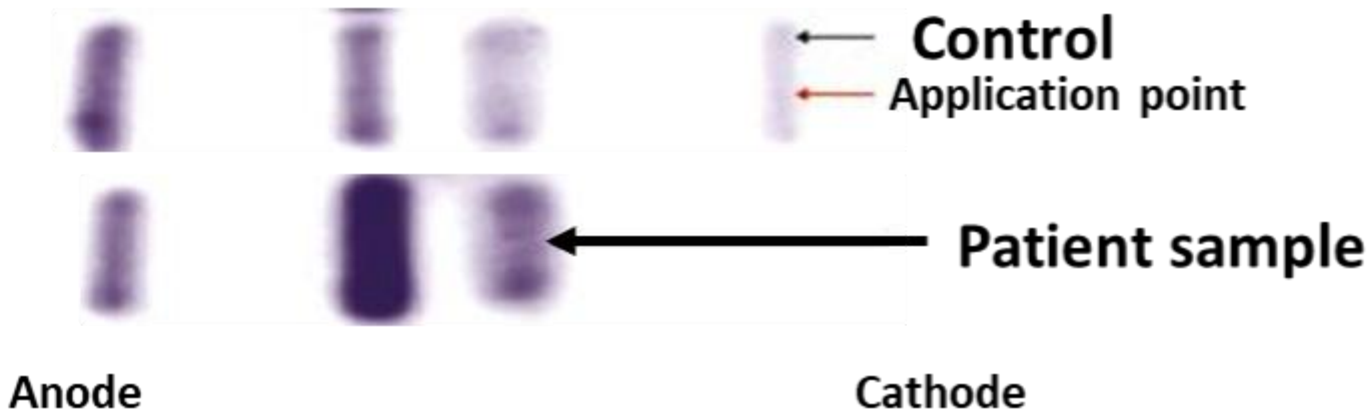
## Serum lipoprotein electrophoresis




SOM.MK.I.BPM1.1.CPR.2.BCHM. 0231; Differentiate different hyperlipidemias (Type I, IIa, IIb, III, IV and V) using lipid profile (Serum TAG and Serum cholesterol), lipoprotein electrophoresis, clinical features and molecular mechanisms and inheritance patterns.

3. A 24-year-old man comes to the physician because of a 1-month history of easy bruising on his arms and legs. He has cystic fibrosis and is non-compliant with his pancreatic enzyme replacement therapy. He denies any trauma, new medications, or alcohol use. His temperature is 37°C (98.6°F), pulse is 85/min, respirations are 18/min, and blood pressure is 122/78 mm Hg. Physical examination shows purplish-blue bruises on the right upper arm and greenish-brown ecchymoses on the lower extremities. Which of the following processes is most likely defective in this patient?

| | | |
|---|---|---|
| **Hemoglobin** | 13.5 g/dL | 13.5-17.5 g/dL |
| **Platelet count** | 350,000/mm$^3$ | 150,000-400,000/mm$^3$ |
| **Bleeding Time** | 6 minutes | 1-9 minutes |
| **PT** | 20 seconds | 11-15 seconds |
| **PTT** | 50 seconds | 25-40 seconds |
| **D-dimer** | 110 ng/mL | </=250 ng/mL |

A. Binding of GPIb to collagen
B. Stabilization of FVIII by von Willebrand factor
C. Carboxylation of glutamic acid residues on clotting factors
D. Fibrinogen binding to GPIIb/IIIa and platelet aggregation
E. Activation of plasminogen and fibrinolysis
F. Hydroxylation of proline and lysine residues in collagen

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SOM.MK.I.BPM1.1.CPR.2.BCHM. 0256 Discuss the role of vitamin K in the post-translational modification of clotting factors

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4. A 65-year-old man comes to the emergency department because of a 30-minute history of severe crushing chest pain radiating to his left arm and jaw. He is given chewable aspirin. He has a 5-year history of hypertension and type 2 diabetes mellitus, both controlled with medications. His blood pressure is 160/90 mmHg, pulse is 104/min, and oxygen saturation is 94% on room air. Physical examination shows diaphoresis and distress. ECG shows ST-segment elevation in leads II, III, and aVF. Serum cardiac biomarkers are requested. Which of the following best explains the rationale behind the administration of the chewed medication?

A. Promotes fibrin breakdown
B. Activates platelet adhesion
C. Prevents platelet aggregation
D. Prevents conversion of a soft clot into a hard clot
E. Prevents activation of Proteins C and S
F. Activates antithrombin-III

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SOM.MK.I.BPM1.1.CPR.2.BCHM. 0260; Indicate the mechanism of action of aspirin, warfarin, Tissue plasminogen activator and heparin

5. A 45-year-old woman comes to the physician because of a 2-day history of fever and loss of appetite. She has had flu-like symptoms four times over the past 7 months and has gained 2.3 kg (5 lb). She has a 2-year history of rheumatoid arthritis and is on corticosteroids for the past 9 months. Her temperature is 37.7°C (100°F), pulse is 100/min, respirations are 20/min, and blood pressure is 130/90 mmHg. Her $O_2$sat: 98% on room air. Physical examination shows a febrile patient with a runny nose and yellow colored mucus. Inhibition of which of the following enzymes best explains the recurrence of her symptoms?

A. Cyclooxygenase-1
B. 5-Lipoxygenase
C. Superoxide dismutase
D. Phospholipase $A_2$
E. NADPH oxidase
F. Glutamate gamma-carboxylase

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SOM.MK.I.BPM1.1.CPR.2.BCHM. 0253; Discuss the role of eicosanoids in the regulation of inflammation (PGE2), gastric acidity, blood flow, and platelet aggregation (PGI2 and TXA2).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0252; Describe the synthesis of the eicosanoid molecules, their general structure, and their precursors. Indicate inhibitors of eicosanoid synthesis.

6. A 15-year-old girl is brought to the physician because of a 6-month history of very heavy bleeding and severe abdominal cramping for the first couple of days during her periods that is relieved by ibuprofen. Menarche was at 13 years, and she has had most cycles with 5-6 days of heavy bleeding with the passage of clots. She has had repeated episodes of nose bleeds in childhood. Laboratory studies show an abnormal ristocetin cofactor assay. Which of the following additional laboratory findings are most likely?

A. Prolonged bleeding time and PT
B. Prolonged bleeding time, PT, and APTT
C. Prolonged PT only
D. Prolonged bleeding time and APTT
E. Prolonged APTT only
F. Increased D-dimer levels
G. Low platelet count

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✔ D. Prolonged bleeding time and APTT
E. Prolonged APTT only
F. Increased D-dimer levels
G. Low platelet count

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0261; Discuss the molecular basis and laboratory findings in the following disorders of hemostasis: Hemophilia (A and B), thrombocytopenia, von Willebrand Disease, Bernard-Soulier Syndrome, Glanzmann thrombasthenia.

# HCB Questions F25

10 Questions

7. A 24-year-old man comes to the emergency department because of a 6-hour history of severe generalized pain, particularly in his lower back and chest, a 1-day history of dark urine, and a 1-week history of feeling fatigued. He has a history of sickle cell disease (HbS genotype), with several prior hospitalizations for similar episodes. His blood pressure is 102/78 mm Hg, respirations are 18/min and temperature is 99.5°F (37.5°C). Physical examination shows an uncomfortable, tachycardic man with mild scleral icterus. Laboratory studies show a hemoglobin of 7.5 g/dL (normal range: 13.5-17.5 g/dL), and elevated levels of indirect bilirubin. A diagnosis of an acute hemolysis during a vaso-occlusive crisis is made. Given the acute hemolysis and his current hemoglobin, which of the following is most likely an additional possible finding on laboratory studies?

A. Decreased total WBC count
B. Elevated reticulocyte count
C. Decreased eosinophil count
D. Neutropenia
E. Decreased basophil count

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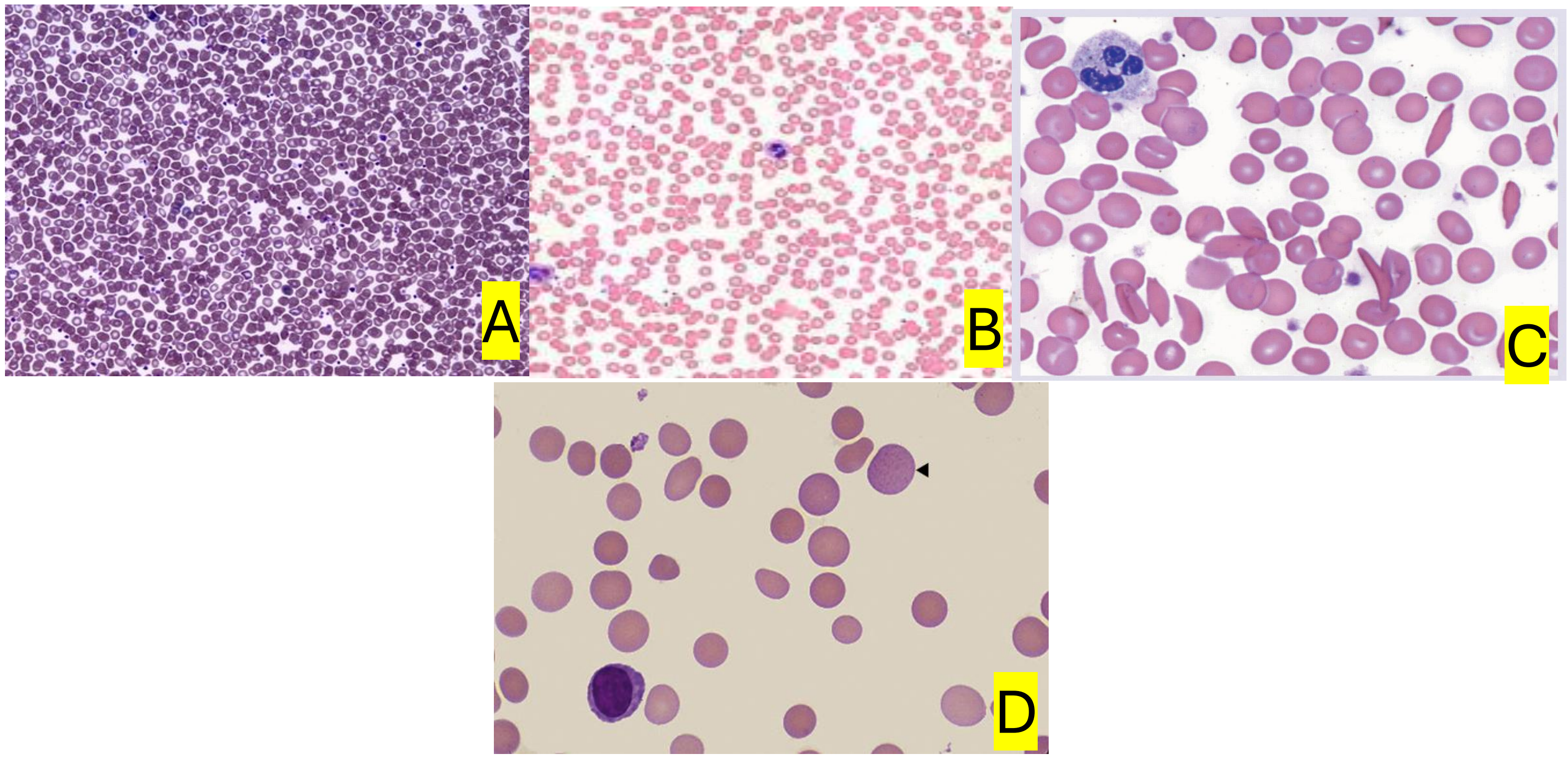
D. Neutropenia
E. Decreased basophil count

SOM.MK.I.BPM1.3.CPR.1.HCB.1179 Describe the process of erythropoiesis.

8. A 10-year-old boy is brought to the physician by his parents because of a 1-week history of fatigue and yellowing of his eyes. His mother reports that he has had similar episodes since early childhood, often following viral illnesses. There is a family history of anemia. Physical examination shows mildly pale and icteric sclera with a palpable spleen 3-cm below the left costal margin, and no lymphadenopathy. Laboratory studies show low hemoglobin, and an elevated reticulocyte count. The Osmotic Fragility Test is positive (RBCs rupture easily when placed in a hypotonic solution). Genetic testing shows a mutation in the gene that codes for the integral membrane protein band 3. Which of the following peripheral smears in the images below best identifies the most likely condition in this patient?

A. A
B. B
C. C
D. D



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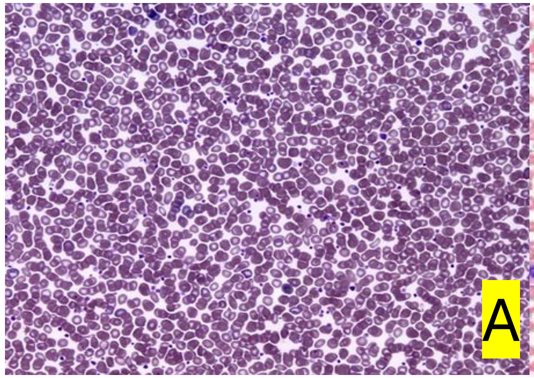

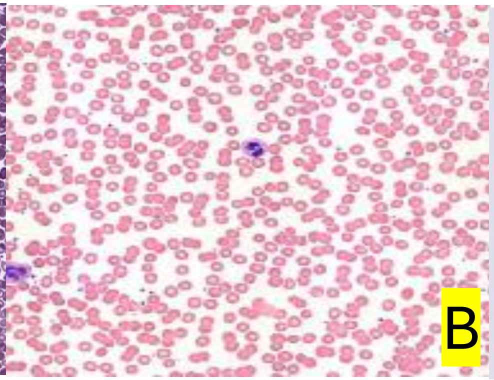

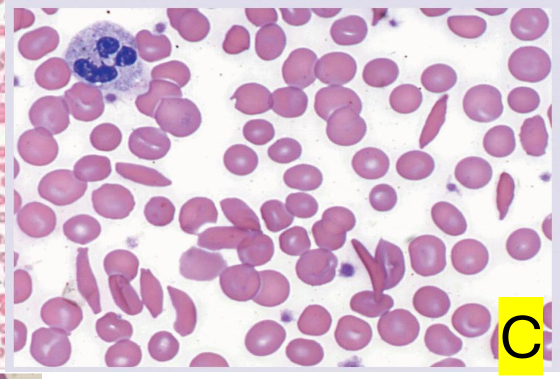

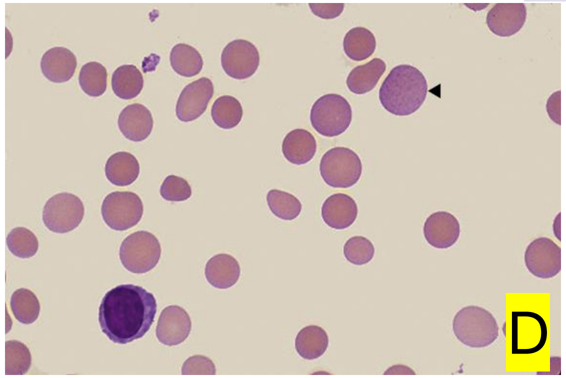


SOM.MK.I.BPM1.3.CPR.1.HCB.1163 Describe the microscopic structure of erythrocytes.

9. A 24-year-old woman comes to the physician because of a 3-month history of persistent fatigue, shortness of breath on exertion, and pale skin. Her medical history is unremarkable, and she denies any bleeding or recent illness. Physical examination shows mild pallor but no jaundice or splenomegaly. Laboratory studies show a normal white blood cell and platelet count, but her red blood cell (RBC) count is slightly decreased. The clinician explains that her symptoms are due to a reduced ability of RBCs to transport oxygen efficiently throughout the body. Which of the following proteins in the membrane of the affected cells most likely helps to bind the protein that carries oxygen?

A. Glycophorin C
B. Band 3
C. Ankyrin complex
D. Band 4.2
E. Spectrin

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10. A 17-year-old girl is brought to the emergency department by her mother because of a 1-week history of bleeding gums, and frequent nosebleeds. Her blood pressure is 118/80 mm Hg, pulse is 86/min, respirations are 22/min and temperature is 98.4°F (36.9 ˚C). Physical examination shows a healed non-bleeding ulceration of the mouth and petechiae across the body. No active bleeding is seen in the nasal cavity. Which of the following characterizes the blood element most likely activated to result in the nasal cavity findings?

A. Derived from the lymphoid lineage
B. Involved in fighting bacterial infections
C. Has an organelle zone rich in granules
D. Has the ability to transport oxygen and carbon dioxide
E. Has cytoplasmic granules that contains major basic protein

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SOM.MK.I.BPM1.3.CPR.1.HCB.1205 Describe the histological structure of thrombocytes.

11. A 75-year-old woman comes to the physician for a follow-up examination one week after surgery.  She reports that she has loss of appetite and has eaten very little in the past week. Physical examination shows a dry, clean wound, and bilateral swollen legs and ankles. The absence of which of the following proteins is most likely responsible for her swollen legs and ankles?

A. Albumin
B. Immunoglobins
C. Fibrinogen
D. Reticular fibers
E. Globulins

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B. Immunoglobins
C. Fibrinogen
D. Reticular fibers
E. Globulins

SOM.MK.I.BPM1.3.CPR.1.HCB.1160 Describe the extracellular components of blood.

12. A 32-year-old woman comes to the physician because of a 2-week history of worsening wheezing, dry cough, and episodic shortness of breath, particularly at night. She has a history of seasonal allergic rhinitis. Physical examination shows diffuse expiratory wheezes and mild nasal congestion. Laboratory studies show an elevated eosinophil count of 12% (normal range: 2-4%). The physician explains the importance of eosinophils in allergic responses. Which of the following best identifies a stage in the normal development of the affected white blood cell in which the specific granules are incorporated?

A. Promyelocyte
B. Myeloblast
C. Common progenitor cell
D. Band cell
E. Myelocyte
F. Metamyelocyte

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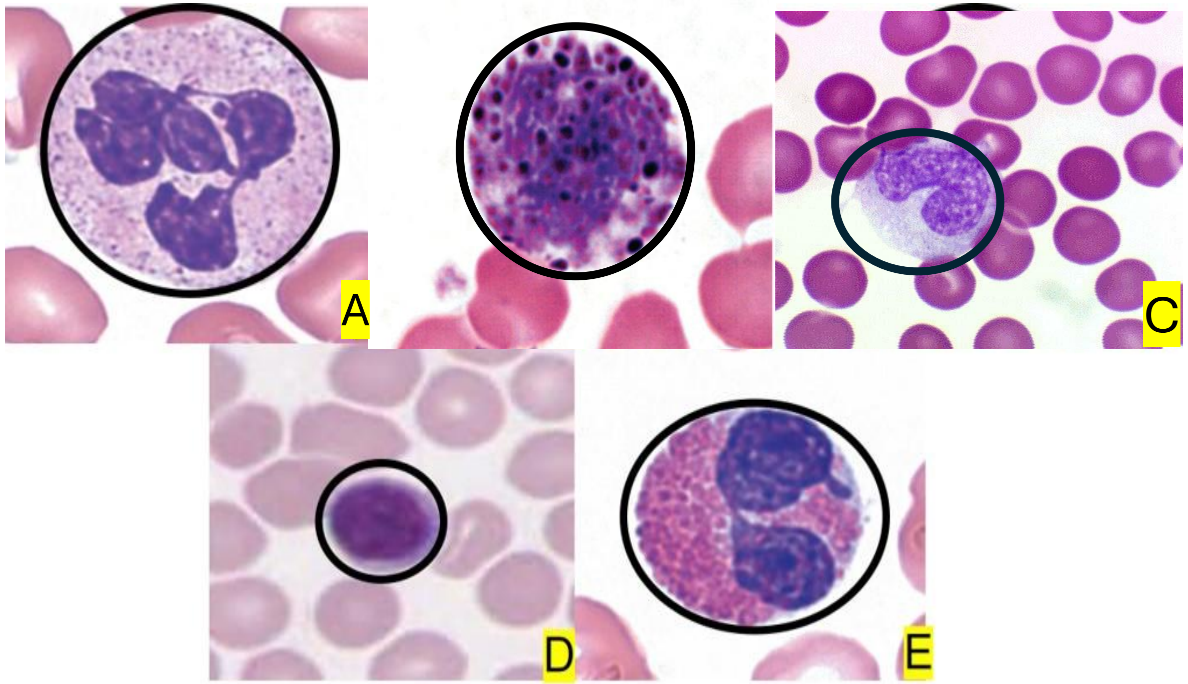
SOM.MK.I.BPM1.3.CPR.1.HCB.1196 Describe the process of granulopoiesis.

13. A 78-year-old man comes to the emergency department because of a 1-day history of fever, chills, poor appetite, shortness of breath and generalized weakness, and a 3-day history of a progressive cough productive of yellow-green sputum. He has a history of COPD. His temperature is 102.2°F (39°C), pulse is 108/min, respirations are 26/min, blood pressure is 138/78 mm Hg, and Spo2 is 90% on room air. Physical examination shows an elderly febrile man in moderate distress with crackles over the right lower lung base. Laboratory studies show an increase in band cells. Which of the following white blood cells in the attached micrographs best characterizes the cell type most likely differentiated from the increased cell?

A. A
B. B
C. C
D. D
E. E



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B. B
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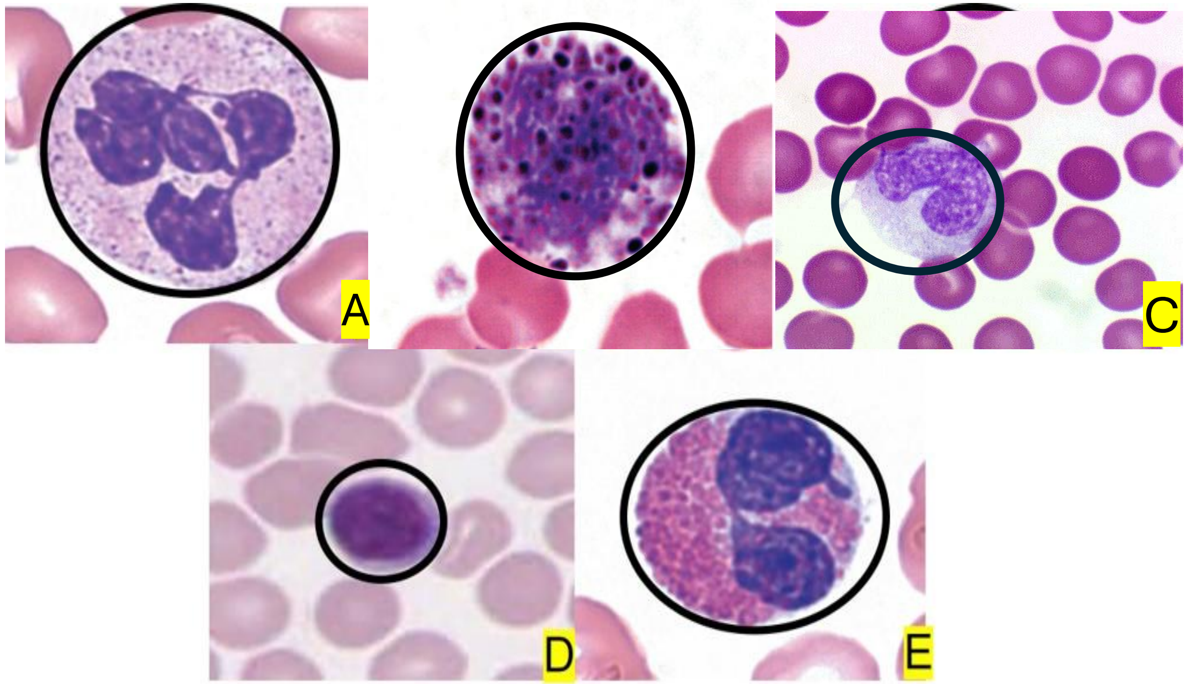


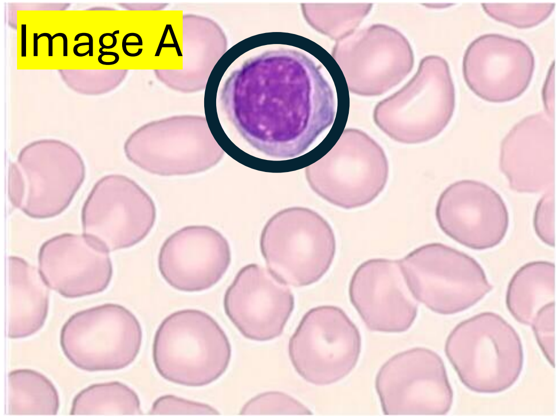

SOM.MK.I.BPM1.3.CPR.1.HCB.1196 Describe the process of granulopoiesis.

14. A 22-year-old woman comes to the emergency department because of a 1-week history of sore throat, low-grade fever, and fatigue. Her temperature is 100.4°F (38°C), pulse is 101/min and blood pressure is 102/78 mm Hg. Physical examination shows tender cervical lymphadenopathy and mild splenomegaly. Laboratory studies show an elevated white blood cell count. The predominantly elevated white blood cell affected is circumscribed in the attached photomicrograph labeled "Image A". Which of the following attached differential white blood cell count panels best defines these findings?

A. Panel A

B. Panel B

C. Panel C




**Panel A**

| Parameter | Result | Reference Range |
|---|---|---|
| WBC | 11.5 ×$10^9$/L | 4.0–10.0 |
| Neutrophils | 3.0 ×$10^9$/L (26%) | 2.0–7.0 (40–75%) |
| Lymphocytes | 7.0 ×$10^9$/L (61%) | 1.0–3.5 (20–45%) |
| Monocytes | 0.8 ×$10^9$/L (7%) | 0.2–1.0 (2–10%) |
| Eosinophils | 0.5 ×$10^9$/L (4%) | 0.0–0.5 (0–6%) |
| Basophils | 0.2 ×$10^9$/L (2%) | 0.0–0.1 (0–1%) |

**Panel B**

| Parameter | Result | Reference Range |
|---|---|---|
| WBC | 14.8 ×$10^9$/L | 4.0–10.0 |
| Neutrophils | 11.8 ×$10^9$/L (80%) | 2.0–7.0 (40–75%) |
| Lymphocytes | 2.0 ×$10^9$/L (14%) | 1.0–3.5 (20–45%) |
| Monocytes | 0.7 ×$10^9$/L (5%) | 0.2–1.0 (2–10%) |
| Eosinophils | 0.2 ×$10^9$/L (1%) | 0.0–0.5 (0–6%) |
| Basophils | 0.1 ×$10^9$/L (0.5%) | 0.0–0.1 (0–1%) |

**Panel C**

| Parameter | Result | Reference Range |
|---|---|---|
| Total WBC Count | 7.2 ×$10^9$/L | 4.0 – 11.0 ×$10^9$/L |
| Neutrophils | 78% (5.6 ×$10^9$/L) | 40 – 70% (2.0 – 7.0 ×$10^9$/L) |
| Lymphocytes | 10% (0.7 ×$10^9$/L) | 20 – 45% (1.0 – 3.0 ×$10^9$/L) |
| Monocytes | 8% (0.6 ×$10^9$/L) | 2 – 10% (0.2 – 0.8 ×$10^9$/L) |
| Eosinophils | 3% (0.2 ×$10^9$/L) | 1 – 6% (0.0 – 0.5 ×$10^9$/L) |
| Basophils | 1% (0.1 ×$10^9$/L) | 0 – 2% (0.0 – 0.2 ×$10^9$/L) |

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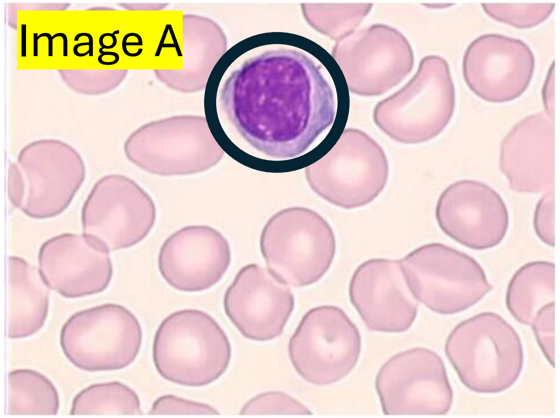
A. Panel A
B. Panel B
C. Panel C




SOM.MK.I.BPM1.3.CPR.1.HCB.1203 Describe the white blood cell count and give an example of clinical conditions that may increase or decrease the cell count for each WBC.

**Panel B**

| Parameter | Result | Reference Range |
|---|---|---|
| WBC | 14.8 $\times 10^9$/L | 4.0–10.0 |
| Neutrophils | 11.8 $\times 10^9$/L (80%) | 2.0–7.0 (40–75%) |
| Lymphocytes | 2.0 $\times 10^9$/L (14%) | 1.0–3.5 (20–45%) |
| Monocytes | 0.7 $\times 10^9$/L (5%) | 0.2–1.0 (2–10%) |
| Eosinophils | 0.2 $\times 10^9$/L (1%) | 0.0–0.5 (0–6%) |
| Basophils | 0.1 $\times 10^9$/L (0.5%) | 0.0–0.1 (0–1%) |

**Panel A**

| Parameter | Result | Reference Range |
|---|---|---|
| WBC | 11.5 $\times 10^9$/L | 4.0–10.0 |
| Neutrophils | 3.0 $\times 10^9$/L (26%) | 2.0–7.0 (40–75%) |
| Lymphocytes | 7.0 $\times 10^9$/L (61%) | 1.0–3.5 (20–45%) |
| Monocytes | 0.8 $\times 10^9$/L (7%) | 0.2–1.0 (2–10%) |
| Eosinophils | 0.5 $\times 10^9$/L (4%) | 0.0–0.5 (0–6%) |
| Basophils | 0.2 $\times 10^9$/L (2%) | 0.0–0.1 (0–1%) |

**Panel C**

| Parameter | Result | Reference Range |
|---|---|---|
| Total WBC Count | 7.2 $\times 10^9$/L | 4.0 – 11.0 $\times 10^9$/L |
| Neutrophils | 78% (5.6 $\times 10^9$/L) | 40 – 70% (2.0 – 7.0 $\times 10^9$/L) |
| Lymphocytes | 10% (0.7 $\times 10^9$/L) | 20 – 45% (1.0 – 3.0 $\times 10^9$/L) |
| Monocytes | 8% (0.6 $\times 10^9$/L) | 2 – 10% (0.2 – 0.8 $\times 10^9$/L) |
| Eosinophils | 3% (0.2 $\times 10^9$/L) | 1 – 6% (0.0 – 0.5 $\times 10^9$/L) |
| Basophils | 1% (0.1 $\times 10^9$/L) | 0 – 2% (0.0 – 0.2 $\times 10^9$/L) |

15. A 45-year-old man comes to the physician for a follow-up examination. He is currently undergoing treatment with high-dose chemotherapy non-Hodgkin lymphoma, diagnosed 2 months ago. Today, he complains of fever, sore throat and oral ulcers. His temperature is 101.3°F (38.5°C). Physical examination shows a febrile, ill-looking man with oral ulcerations. The laboratory studies are attached for your reference. Which of the following hematopoietic growth factors will most likely stimulate the growth and differentiation of progenitors of the affected cell type?

A. Thrombopoietin
B. Nitric oxide
C. Erythropoietin
D. Endothelin
E. Colony stimulating-factor

| **Test** | **Result** | **Reference Range** |
|---|---|---|
| WBC count | 900 /μL | 4,000–11,000 /μL |
| Neutrophils | 10% | 40–70% |
| Lymphocytes | 75% | 20–45% |
| Monocytes | 10% | 2–8% |
| Eosinophils | 4% | 1–4% |
| Basophils | 1% | <1% |
| Hemoglobin | 13.8 g/dL | 13.5–17.5 g/dL |
| Platelet count | 250,000 /μL | 150,000–400,000 /μL |

15. A 45-year-old man comes to the physician for a follow-up examination. He is currently undergoing treatment with high-dose chemotherapy non-Hodgkin lymphoma, diagnosed 2 months ago. Today, he complains of fever, sore throat and oral ulcers. His temperature is 101.3°F (38.5°C). Physical examination shows a febrile, ill-looking man with oral ulcerations. The laboratory studies are attached for your reference. Which of the following hematopoietic growth factors will most likely stimulate the growth and differentiation of progenitors of the affected cell type?

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| **Test** | **Result** | **Reference Range** |
|---|---|---|
| WBC count | 900 /µL | 4,000–11,000 /µL |
| Neutrophils | 10% | 40–70% |
| Lymphocytes | 75% | 20–45% |
| Monocytes | 10% | 2–8% |
| Eosinophils | 4% | 1–4% |
| Basophils | 1% | <1% |
| Hemoglobin | 13.8 g/dL | 13.5–17.5 g/dL |
| Platelet count | 250,000 /µL | 150,000–400,000 /µL |

SOM.MK.I.BPM1.3.CPR.1.HCB. Describe the role of growth factors, cytokines and hormones involved in hematopoiesis.

16. A 30-year-old woman is bought to the emergency department because of a 2-hour history of hives, facial swelling, and wheezing shortly after eating shellfish. She also reports occasional nasal congestion and itchy eyes. Physical examination shows urticarial rash and mild facial edema. Laboratory studies show an elevated basophil count. Enzyme immunoassay shows elevated levels of histamine. Which of the following best identifies a function of the elevated protein on enzyme assay?

A. Promotes vasodilation
B. Anticoagulation
C. Vasoconstriction
D. Bronchoconstriction
E. Production of antibodies

16. A 30-year-old woman is bought to the emergency department because of a 2-hour history of hives, facial swelling, and wheezing shortly after eating shellfish. She also reports occasional nasal congestion and itchy eyes. Physical examination shows urticarial rash and mild facial edema. Laboratory studies show an elevated basophil count. Enzyme immunoassay shows elevated levels of histamine. Which of the following best identifies a function of the elevated protein on enzyme assay?

A. Promotes vasodilation
B. Anticoagulation
C. Vasoconstriction
D. Bronchoconstriction
E. Production of antibodies

SOM.MK.I.BPM1.3.CPR.1.HCB.1195 Discuss how the products secreted from the specific and non specific granules of basophils aid in their functions